A Corrected Partial Analysis of Sequential Issue-by-Issue Voting Without LAMO

Prepared note

June 24, 2026

Abstract

This note revisits a proposed solution attempt for the open problem of Gershkov, Moldovanu, and Shi asking whether failure of left-additive mutual orthogonality (LAMO) of the issue basis forces order dependence in sequential issue-by-issue voting. The full conjecture is *not* proved here. The main unconditional result below is a sharp local characterization: LAMO is exactly the condition under which sincere voting is always optimal at a final issue. Thus every non-LAMO basis creates a strictly manipulable final-stage continuation. I then give a rigorous higher-dimensional extension of the known two-dimensional converse under a checkable coordinate-plane hypothesis: the non-LAMO failure occurs inside a two-dimensional coordinate plane and all remaining issues form a LAMO tail. This is genuine partial progress, but it does not remove the principal obstacle in the authors' problem: in dimensions at least three a non-LAMO witness may be an additive combination of several issues, and the separate sequential votes on those issues need not behave like one block issue. A Hadamard example in ℓ_4^4 verifies this additive obstruction even for a pairwise BJ-mutually orthogonal basis.

Status of the result

The conclusion of this note is deliberately modest. It gives a corrected, rigorous partial analysis. It does **not** solve the general open problem. The serious overclaims in the original attempt have been removed: in particular, the block-localization reduction is not asserted to be an equivalence, and no theorem is claimed for arbitrary higher-dimensional non-LAMO bases.

1 The open problem

Let $(X, \|\cdot\|)$ be a finite-dimensional real normed space, identified with $\mathbb{R}^d$, and let

$$B = (x_1, \ldots, x_d)$$

be an algebraic basis. Every alternative $v \in X$ has unique coordinates

$$v = \sum_{j=1}^{d} \alpha_j(v) x_j.$$

In the sequential issue-by-issue voting procedure associated with an order $\sigma = (\sigma_1, \ldots, \sigma_d)$, the coordinate along x_{σ_1} is chosen first, then the coordinate along x_{σ_2}, and so on. At each stage voters submit real reports and the stage outcome is the median of the reports. Voters have norm-induced spatial preferences with peaks t_i: voter i weakly prefers v to v' if and only if

$$\|t_i - v\| \leq \|t_i - v'\| .$$

We use the same equilibrium-selection convention as Gershkov, Moldovanu, and Shi: the outcome selected by iteratively undominated ex-post perfect equilibrium play.

For vectors $u, v \in X$, write

$$u \dashv v \quad \Longleftrightarrow \quad \|u + \lambda v\| \geq \|u\| \quad \text{for every } \lambda \in \mathbb{R}.$$

This is Birkhoff-James orthogonality. For a subspace $Y \subseteq X$, write

$$Y \dashv v \quad \Longleftrightarrow \quad y \dashv v \quad \text{for every } y \in Y.$$

For each j, set

$$X_j = \operatorname{span}\{x_k : k \neq j\}.$$

The basis B is *left-additively mutually orthogonal*, or *LAMO*, if

$$X_j \dashv x_j \qquad \text{for every } j = 1, \ldots, d.$$

Equivalently, every linear combination of all basis vectors other than x_j is Birkhoff-James orthogonal to x_j. Thus B is not LAMO precisely when there exist j, $y \in X_j$, and $\lambda \in \mathbb{R}$ such that

$$\|y + \lambda x_j\| < \|y\| . \tag{1}$$

Gershkov, Moldovanu, and Shi prove that LAMO implies order independence: for any order, equilibrium voting is sincere and the sequential outcome coincides with the static coordinatewise median. They also prove the converse in dimension two. The general open problem asks whether

$$\neg\text{LAMO} \quad \Longrightarrow \quad \exists \sigma, \pi, \exists (t_1, \ldots, t_n) : O_\sigma(t_1, \ldots, t_n) \neq O_\pi(t_1, \ldots, t_n),$$

where O_σ denotes the selected equilibrium outcome for order σ.

2 The exact last-stage meaning of LAMO

The following lemma is the main unconditional statement. It is elementary, but it is exactly where LAMO enters the backward-induction proof of order independence.

Lemma 2.1 (Last-stage projection test). *Fix an issue j. Suppose all coordinates except the coordinate along x_j have already been fixed at a vector $w \in X_j$, and suppose voter i's peak is*

$$t_i = u + a x_j, \qquad u \in X_j, \quad a \in \mathbb{R}.$$

If voter i is pivotal at the last stage, then the set of optimal reports is

$$a + P_j(w - u), \qquad P_j(z) := \operatorname*{arg\,min}_{r \in \mathbb{R}} \|z + r x_j\| .$$

In particular, truthful last-stage reporting is optimal for every possible history $w \in X_j$ and every possible peak $u + a x_j$ if and only if $X_j \dashv x_j$.

Proof. If voter i reports $\widehat{a}$, the final alternative is $w + \widehat{a}x_j$. The distance from voter i's peak is

$$\|w + \widehat{a}x_j - (u + ax_j)\| = \|w - u + (\widehat{a} - a)x_j\|.$$

Thus $\widehat{a}$ is optimal exactly when $\widehat{a} - a \in P_j(w - u)$, proving the first assertion.

Truthful reporting means $\widehat{a} = a$, hence means $0 \in P_j(w - u)$. Since $w - u$ ranges over all of X_j, truthful reporting is optimal for all last-stage histories and all peaks if and only if

$$\|z + rx_j\| \geq \|z\| \qquad \text{for every } z \in X_j \text{ and every } r \in \mathbb{R},$$

which is exactly $X_j \dashv x_j$. □

Corollary 2.2 (Non-LAMO gives a strict local manipulation). *If B is not LAMO, then there exist an issue j, a last-stage history $w \in X_j$, a voter peak t, and a report $\widehat{a} \neq \alpha_j(t)$ such that a pivotal voter strictly prefers the outcome induced by $\widehat{a}$ to the outcome induced by truthful reporting.*

Proof. Since B is not LAMO, there exist j, $y \in X_j$, and $\lambda \in \mathbb{R}$ satisfying (1). In Lemma 2.1, take $w = y$, $u = 0$, and $a = 0$. Truthful reporting gives distance $\|y\|$, whereas reporting λ gives the strictly smaller distance $\|y + \lambda x_j\|$. □

Remark 2.3 (Why Corollary 2.2 is not the conjecture). *Corollary 2.2 is a local continuation statement. It does not say that the history $w = y$ is reached in equilibrium, nor does it say that the same profile produces a different outcome under another order. This is precisely the higher-dimensional difficulty: a deviation that is profitable at the final issue may be anticipated and offset by strategic behavior on previous or later issues in another order.*

3 A genuine partial theorem: a non-LAMO coordinate plane with a LAMO tail

The authors' two-dimensional converse is the main positive tool available. In dimension two, LAMO is just mutual Birkhoff-James orthogonality of the two issue directions, and failure of this property implies order dependence. The following theorem gives a precise condition under which that two-dimensional result embeds into a higher-dimensional problem.

Theorem 3.1 (Coordinate-plane reduction with a LAMO tail). *Let $j \neq k$, and set*

$$S = \operatorname{span}\{x_j, x_k\}, \qquad H = \operatorname{span}\{x_\ell : \ell \notin \{j, k\}\}.$$

Assume:

(A1) *The restricted two-dimensional basis (x_j, x_k) is not LAMO in the normed plane $(S, \|\cdot\||_S)$.*

(A2) *Every issue outside the plane is left-additively orthogonal to its complement in the full space:*

$$X_\ell \dashv x_\ell \qquad \text{for every } \ell \notin \{j, k\}.$$

Then the authors' order-dependence conclusion holds for B: there are two voting orders and a peak profile for which the corresponding selected iteratively undominated ex-post perfect equilibrium outcomes differ.

Proof. Let $R = (\ell_1, \dots, \ell_{d-2})$ be any ordering of the indices outside $\{j, k\}$. Consider the two full orders

$$\sigma = (j, k, \ell_1, \dots, \ell_{d-2}), \qquad \pi = (k, j, \ell_1, \dots, \ell_{d-2}).$$

Restrict attention to profiles $P = (t_1, \dots, t_n)$ with all peaks in S.

We first claim that, after the first two stages of either order, the remaining H-issues are voted truthfully and produce the zero H-coordinate. Fix any continuation after the first two stages. The already determined S-part is some $s \in S$. Consider a tail issue x_{ℓ_r}. Suppose, inductively from the end of the tail, that all later tail issues are reported truthfully and hence have outcome zero. Let the earlier tail outcomes contribute some vector

$$h \in \operatorname{span}\{x_{\ell_1}, \dots, x_{\ell_{r-1}}\}.$$

For voter i, whose ideal point lies in S, a pivotal report q on issue x_{ℓ_r} gives final residual

$$z + q x_{\ell_r}, \qquad z = s - t_i + h.$$

The vector z has no x_{ℓ_r}-coordinate, so $z \in X_{\ell_r}$. By assumption (A2),

$$\|z\| \le \|z + q x_{\ell_r}\| \qquad \forall q \in \mathbb{R}.$$

Thus the truthful coordinate $q = 0$ is a best response at this tail stage. Backward induction over $r = d-2, d-3, \dots, 1$ shows that the whole tail has the same iteratively undominated continuation as a LAMO sequential median: all voters report their true tail coordinates, which are zero, and the tail outcome is zero.

It follows that the strategic problem in the first two stages is exactly the two-dimensional sequential voting game on $(S, \|\cdot\| \,|_S)$ with issue basis (x_j, x_k), respectively (x_k, x_j). Indeed, every peak lies in S and the tail continuation contributes no H-component, so voter i's final loss after the first two coordinates produce $s' \in S$ is simply $\|s' - t_i\|$, the same loss as in the two-dimensional restricted game.

By assumption (A1) and the two-dimensional converse theorem of Gershkov, Moldovanu, and Shi, there is an odd number of voters and a profile $P \subset S$ for which the selected equilibrium outcomes of the two-dimensional games with orders (x_j, x_k) and (x_k, x_j) differ. For this same profile, the full d-dimensional games with orders σ and π reproduce those two-dimensional outcomes in S and have zero H-component. Hence the full outcomes differ. □

Corollary 3.2 (Pairwise failure isolated from a LAMO tail)**.** *Suppose B fails LAMO because of a pair of basis vectors, in the sense that for some $j \neq k$ either*

$$\operatorname{span}\{x_k\} \not\dashv x_j \quad \text{or} \quad \operatorname{span}\{x_j\} \not\dashv x_k,$$

and suppose $X_\ell \dashv x_\ell$ for every $\ell \notin \{j, k\}$. Then sequential issue-by-issue voting with respect to B is order dependent.

Proof. The first hypothesis says exactly that the restricted two-dimensional basis (x_j, x_k) is not LAMO. Theorem 3.1 applies. □

Example 3.3 (A simple family covered by Theorem 3.1)**.** *Let $X = (\mathbb{R}^d, \|\cdot\|_p)$ with $1 < p < \infty$, and let*

$$x_1 = e_1, \qquad x_2 = e_1 + e_2, \qquad x_m = e_m \quad (3 \le m \le d).$$

In the plane $S = \operatorname{span}\{x_1, x_2\} = \operatorname{span}\{e_1, e_2\}$, the pair (x_1, x_2) is not LAMO. Indeed, the differentiable ℓ_p orthogonality test gives

$$x_1 \dashv x_2 \quad \iff \quad \sum_r |(x_1)_r|^{p-2}(x_1)_r(x_2)_r = 0,$$

and the sum equals 1. For each $m \geq 3$, every vector in X_m has zero mth standard coordinate, so

$$\|y + \lambda e_m\|_p^p = \|y\|_p^p + |\lambda|^p \geq \|y\|_p^p \qquad (y \in X_m).$$

Thus the remaining issues form a LAMO tail, and Theorem 3.1 proves order dependence.

4 A corrected conditional block reduction

A tempting argument is to take a non-LAMO witness $y \in X_j$ from (1) and reduce the problem to the two-dimensional plane

$$S = \operatorname{span}\{y, x_j\}.$$

On this plane, the ordered pair (y, x_j) is not LAMO, so the known two-dimensional converse would imply order dependence. The issue is that y need not be one of the original issues; it may be a linear combination of several issues, and the original mechanism votes on those coordinates separately.

The following proposition is the valid conditional form of this idea. It is not an equivalence and should not be read as proving that all non-LAMO failures localize.

Definition 4.1 (Two-dimensional equilibrium localization)**.** *Let $y \in X_j \setminus \{0\}$ and set $S = \operatorname{span}\{y, x_j\}$. We say that the non-LAMO witness (y, x_j) is* equilibrium-localizable *if there are two full voting orders σ, π of the original basis B such that, for every odd electorate size n and every profile $(t_1, \ldots, t_n) \in S^n$, the selected equilibrium outcomes of the original d-issue games under σ and π both lie in S and agree with the selected equilibrium outcomes of the corresponding two-issue games on S with issue basis (y, x_j), in the two opposite orders.*

Proposition 4.2 (Conditional two-dimensional reduction)**.** *Suppose B is not LAMO and has a non-LAMO witness (y, x_j) that is equilibrium-localizable in the sense of Definition 4.1. Then sequential issue-by-issue voting with respect to B is order dependent.*

Proof. Because $\|y + \lambda x_j\| < \|y\|$ for some λ, the basis (y, x_j) of the two-dimensional space S is not LAMO. By the known two-dimensional converse of Gershkov, Moldovanu, and Shi, there is a profile in S^n for which the two orders of (y, x_j) produce different selected equilibrium outcomes. By equilibrium-localizability, the two full orders σ and π in the original game reproduce those two distinct outcomes. Hence the original d-issue game is order dependent. □

The entire unresolved part is hidden in Definition 4.1. For a general non-LAMO witness, no proof is given here that the separate votes on the basis coordinates forming y behave like a single vote on y.

5 A verified additive obstruction

The following example strengthens the illustrative example in the original attempt. It shows that even pairwise mutual Birkhoff-James orthogonality of basis vectors need not imply LAMO.

Example 5.1 (A pairwise BJ-mutually orthogonal basis that is not LAMO)**.** *Let* $X = (\mathbb{R}^4, \|\cdot\|_4)$*, where*

$$\|x\|_4 = \left(\sum_{r=1}^{4} |x_r|^4\right)^{1/4}.$$

Define

$$h_1 = (1,1,1,1), \qquad h_2 = (1,1,-1,-1),$$
$$h_3 = (1,-1,1,-1), \qquad h_4 = (1,-1,-1,1).$$

Then (h_1, h_2, h_3, h_4) *is pairwise BJ-mutually orthogonal, but it is not LAMO.*

Proof. For $1 < p < \infty$, the function $\lambda \mapsto \|a + \lambda b\|_p^p$ is convex and differentiable, and

$$\frac{d}{d\lambda}\bigg|_{\lambda=0} \|a + \lambda b\|_p^p = p \sum_{r=1}^{m} |a_r|^{p-2} a_r b_r.$$

Thus, in ℓ_p^m,

$$a \dashv b \quad \Longleftrightarrow \quad \sum_{r=1}^{m} |a_r|^{p-2} a_r b_r = 0.$$

For $p = 4$, if a has all coordinates equal to ± 1, then $|a_r|^2 a_r = a_r$. The four vectors h_1, h_2, h_3, h_4 are the rows of a Hadamard matrix, so their Euclidean dot products are zero off the diagonal. Therefore

$$h_i \dashv h_j \qquad \text{for all } i \neq j,$$

and the basis is pairwise BJ-mutually orthogonal.

Now set

$$y = h_2 + h_3 + h_4 = (3, -1, -1, -1) \in \operatorname{span}\{h_2, h_3, h_4\} = X_1.$$

For $p = 4$,

$$\sum_{r=1}^{4} |y_r|^2 y_r (h_1)_r = 27 - 1 - 1 - 1 = 24 \neq 0.$$

Hence $y \not\dashv h_1$, so $X_1 \not\dashv h_1$. Therefore the basis is not LAMO.

Equivalently, if

$$f(\lambda) = \|y + \lambda h_1\|_4^4,$$

then $f'(0) = 4 \cdot 24 > 0$. Hence for all sufficiently small negative λ,

$$\|y + \lambda h_1\|_4 < \|y\|_4.$$

This is an explicit additive non-LAMO witness. $\square$

Remark 5.2. *Example 5.1 explains why a proof based only on two-coordinate restrictions cannot settle the general conjecture. Every pair of basis vectors passes the mutual BJ test, but the hyperplane spanned by three of them is not BJ-orthogonal to the remaining vector. The failure is genuinely additive.*

6 What has been fixed relative to the original attempt

The corrected conclusions are as follows.

1. The last-stage lemma is valid and is now stated as an exact equivalence: LAMO is precisely the condition that truthful final-issue reporting is always optimal at every last-stage continuation.
2. Non-LAMO gives a strict local manipulation, but this is not the same as order dependence. The missing step is to put the relevant continuation on an equilibrium path and to show that another order gives a different selected outcome.
3. The two-dimensional reduction is valid only under an explicit localization hypothesis. It is not an equivalence, and the note does not prove that every non-LAMO witness is localizable.
4. The coordinate-plane theorem gives a genuine, verified class of higher-dimensional partial results. It proves order dependence when a non-LAMO pair is isolated from the remaining issues by a LAMO tail.
5. The additive Hadamard example shows that pairwise mutual BJ orthogonality is not enough for LAMO, so any proof of the full conjecture must control linear combinations of previously or later voted issues, not merely pairs of issues.

7 Conclusion

This note obtains partial progress, not a solution. The full conjecture remains unproved here:

$$B \text{ not LAMO} \quad \overset{?}{\Longrightarrow} \quad \text{sequential issue-by-issue voting with respect to } B \text{ is order dependent.}$$

What is proved is that non-LAMO is exactly the obstruction to universal truthful last-stage behavior, and that the known two-dimensional converse embeds cleanly when the non-LAMO plane is followed by a LAMO tail. The unresolved mathematical task is to convert an arbitrary additive non-LAMO witness $y \in X_j$ into an equilibrium-path phenomenon for the original separate coordinate votes. Example 5.1 shows why this task is genuinely higher-dimensional.

References

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